Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Manuel & Jeanette Borges Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
10408 S Van Allen RD	Stockton	San Joaquin	95215

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1920

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X185-X110-XX05-XXXX X205-X180-XX03-XXXX

B. OPERATORS

Borges, Manuel

Operator name: Borges, Manuel

Telephone no.: (209) 466-1828 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
18355 Munro DR	Stockton	CA	95215

This operator is responsible for paying permit fees.

C. OWNERS

Borges, Jeannette

Legal owner name: Borges, Jeannette

Telephone no.: (209) 466-1828 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
18355 Munro DR	Stockton	CA	95215

Borges, Manuel

Legal owner name:	Borges, Manuel	Telephone no.:	(209) 466-1828	
			Landline	Cellular
18355 Munro DR		Stockton	CA	95215
Mailing Address Number and Street		City	State	Zip Code

This owner is responsible for paying permit fees.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	250	525	340	195	100
Number under roof	1,070	0	0	0	0	0
Maximum number	1,100	250	695	340	195	165
Average number	1,100	250	675	340	195	150
Avg live weight (lbs)	1,400	1,450	950	700		

Predominant milk cow breed: Holstein

Average milk production: 74 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 43,749.53 *tons per reporting period*

Total nitrogen from manure: 535,247.73 *lbs per reporting period*

After ammonia losses (30% loss applied): 374,673.41 *lbs per reporting period*

Total phosphorus from manure: 87,267.11 *lbs per reporting period*

Total potassium from manure: 213,550.95 *lbs per reporting period*

Total salt from manure: 575,422.50 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 30,492,000 *gallons*

Total nitrogen generated: 107,270.99 *lbs*

Total phosphorus generated: 20,717.82 *lbs*

Total potassium generated: 176,536.93 *lbs*

Total salt generated: 1,221,266.38 *lbs*

	30,492,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	30,492,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Ground water
IW 1	Ground water
IW 2	Ground water
IW 3	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/05/2023	Solid commercial fertilizer *4-0-0*	32.40 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
06/27/2023	Solid commercial fertilizer *UN 32*	5.65 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	6,201.79	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	6,201.79	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Big Field	51	51	2	both	X205-X180-XX02-XXXX
Faria 1	28	28	2	process wastewater	X205-X180-XX02-XXXX
Faria 2	21	21	2	process wastewater	X205-X180-XX02-XXXX
Morris	50	50	2	both	X185-X110-XX05-XXXX
Skiff	41	41	2	manure	X205-X180-XX02-XXXX
The 40	21	21	2	both	X205-X180-XX02-XXXX X205-X180-XX03-XXXX
The 80	65	65	2	process wastewater	X205-X180-XX02-XXXX
Tules	47	47	2	process wastewater	X205-X180-XX02-XXXX
Totals for areas that were used for application	324	324	16		
Totals for areas that were not used for application					
Land application area totals	324	324	16		

B. CROPS AND HARVESTS

Big Field

Field name: Big Field

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 51 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	995.05 *ton*	As-is		64.2	4,760.00	1,500.00	16,500.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.51	185.74	58.53	643.86	1,522.70

Big Field

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 51 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/03/2023	1,436.93 *ton*	As-is		66.4	4,130.00	870.00	4,600.00		5.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.18	232.73	49.02	259.21	1,003.48

Faria 1

Field name: Faria 1

10/28/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 28 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	546.28 *ton*	As-is		70.0	4,410.00	1,200.00	15,300.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.51	172.08	46.82	597.01	1,264.25

06/02/2023: Corn, silage

Crop: Corn, silage Acres planted: 28 Plant date: 06/02/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	789.60 *ton*	As-is		67.4	3,680.00	1,240.00	5,400.00		2.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.20	207.55	69.94	304.56	422.89

Faria 2

Field name: Faria 2

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	409.71 *ton*	As-is		77.6	4,410.00	1,200.00	15,300.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.51	172.08	46.82	597.01	943.97

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	592.20 *ton*	As-is		67.2	3,680.00	1,240.00	5,400.00		2.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.20	207.55	69.94	304.56	425.48

Morris

Field name: Morris

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 50 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	864.00 *ton*	As-is		72.4	3,530.00	880.00	13,200.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	17.28	122.00	30.41	456.19	1,087.40

Morris

06/11/2023: Corn, silage

Crop: Corn, silage Acres planted: 50 Plant date: 06/11/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	1,023.00 *ton*	As-is		61.4	3,740.00	860.00	5,650.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	20.46	153.04	35.19	231.20	1,200.43

Skiff

Field name: Skiff

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 41 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/04/2023	799.91 *ton*	As-is		70.0	2,880.00	1,080.00	12,600.00		9.99

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.51	112.38	42.14	491.65	1,169.43

05/31/2023: Corn, silage

Crop: Corn, silage Acres planted: 41 Plant date: 05/31/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/29/2023	1,156.20 *ton*	As-is		63.4	4,500.00	1,270.00	6,390.00		3.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.20	253.80	71.63	360.40	805.05

The 40

Field name: The 40

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	409.71 *ton*	As-is		72.0	3,740.00	750.00	6,740.00		8.69

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.51	145.93	29.27	262.99	949.43

06/06/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	592.20 *ton*	As-is		66.4	4,800.00	990.00	5,030.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.20	270.72	55.84	283.69	1,231.78

The 80

Field name: The 80

11/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 65 Plant date: 11/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	1,268.15 *ton*	As-is		76.6	4,240.00	1,030.00	13,100.00		7.31

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.51	165.44	40.19	511.16	667.45

The 80

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 65 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/05/2023	1,833.00 *ton*	As-is		67.0	3,730.00	900.00	5,900.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.20	210.37	50.76	332.76	1,265.62

Tules

Field name: Tules

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/05/2023	916.97 *ton*	As-is		79.2	4,110.00	1,290.00	15,600.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	19.51	160.37	50.34	608.71	941.47

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/30/2023	1,325.40 *ton*	As-is		69.6	3,620.00	1,390.00	6,280.00		2.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.20	204.17	78.40	354.19	445.79

NUTRIENT BUDGET

A. LAND APPLICATIONS

Big Field - 10/28/2022: Oats, silage-soft dough

Field name: Big Field

Crop: Oats, silage-soft dough — Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	142.55	0.76	163.29	1,563.75	2,640,000.00 *gal*
IW 1	Ground water	0.00	0.00	0.00	84.24	2,640,000.00 *gal*
Application event totals		142.55	0.76	163.29	1,647.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.66	22.81	167.78	901.97	1,056,000.00 *gal*
Application event totals		98.66	22.81	167.78	901.97	

Big Field - 06/05/2023: Corn, silage

Field name: Big Field

Crop: Corn, silage — Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	275.33	131.68	710.77	6,718.62	690.00 *ton*
Application event totals		275.33	131.68	710.77	6,718.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	80.86	6,678,000.00 *gal*
Application event totals		0.00	0.00	0.00	80.86	

Big Field - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	76.28	6,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	76.28	

06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	73.23	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	73.23	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	76.28	6,300,000.00 *gal*
Application event totals		0.00	0.00	0.00	76.28	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	73.23	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	73.23	

Big Field - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	70.18	5,796,000.00 *gal*
Application event totals		0.00	0.00	0.00	70.18	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	67.13	5,544,000.00 *gal*
Application event totals		0.00	0.00	0.00	67.13	

Faria 1 - 10/28/2022: Oats, silage-soft dough

Field name: Faria 1

Crop: Oats, silage-soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.77	0.00	0.00	194.74	2,970,000.00 *gal*
Application event totals		1.77	0.00	0.00	194.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.32	25.96	191.00	1,026.79	660,000.00 *gal*
Application event totals		112.32	25.96	191.00	1,026.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.32	25.96	191.00	1,026.79	660,000.00 *gal*
Application event totals		112.32	25.96	191.00	1,026.79	

Faria 1 - 06/02/2023: Corn, silage

Field name: Faria 1

Crop: Corn, silage

Plant date: 06/02/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	80.59	3,654,000.00 *gal*
Application event totals		0.00	0.00	0.00	80.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/02/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	77.81	3,528,000.00 *gal*
Application event totals		0.00	0.00	0.00	77.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	133.05	32.89	216.02	1,442.62	1,254,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	52.80	2,394,000.00 *gal*
Application event totals		133.05	32.89	216.02	1,495.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	72.25	3,276,000.00 *gal*
Application event totals		0.00	0.00	0.00	72.25	

Faria 1 - 06/02/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	126.05	31.16	204.65	1,366.70	1,188,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	50.02	2,268,000.00 *gal*
Application event totals		126.05	31.16	204.65	1,416.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	72.25	3,276,000.00 *gal*
Application event totals		0.00	0.00	0.00	72.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	69.47	3,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	69.47	

Faria 2 - 10/29/2022: Oats, silage-soft dough

Field name: Faria 2

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.84	0.00	0.00	201.95	2,310,000.00 *gal*
Application event totals		1.84	0.00	0.00	201.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	112.32	25.96	191.00	1,026.79	495,000.00 *gal*
Application event totals		112.32	25.96	191.00	1,026.79	

Faria 2 - 10/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	104.83	24.23	178.27	958.34	462,000.00 *gal*
Application event totals		104.83	24.23	178.27	958.34	

Faria 2 - 06/04/2023: Corn, silage

Field name: Faria 2

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	81.51	2,772,000.00 *gal*
Application event totals		0.00	0.00	0.00	81.51	

06/04/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	77.81	2,646,000.00 *gal*
Application event totals		0.00	0.00	0.00	77.81	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	130.72	32.31	212.23	1,417.31	924,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	51.87	1,764,000.00 *gal*
Application event totals		130.72	32.31	212.23	1,469.19	

Faria 2 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	77.81	2,646,000.00 *gal*
Application event totals		0.00	0.00	0.00	77.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	121.38	30.00	197.07	1,316.08	858,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	48.17	1,638,000.00 *gal*
Application event totals		121.38	30.00	197.07	1,364.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.10	2,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	70.40	2,394,000.00 *gal*
Application event totals		0.00	0.00	0.00	70.40	

Morris - 10/30/2022: Oats, silage-soft dough

Field name: Morris

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	80.99	38.72	159.84	962.59	300.00 *ton*
Application event totals		80.99	38.72	159.84	962.59	

Morris - 10/30/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	0.00	0.00	0.00	176.14	5,148,000.00 *gal*
Application event totals		0.00	0.00	0.00	176.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.43	9.91	160.47	1,321.85	1,584,000.00 *gal*
Application event totals		81.43	9.91	160.47	1,321.85	

Morris - 06/11/2023: Corn, silage

Field name: Morris

Crop: Corn, silage

Plant date: 06/11/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	80.92	6,552,000.00 *gal*
Application event totals		0.00	0.00	0.00	80.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/11/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	76.25	6,174,000.00 *gal*
Application event totals		0.00	0.00	0.00	76.25	

Morris - 06/11/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	212.65	14.21	347.51	3,358.82	2,178,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	51.35	4,158,000.00 *gal*
Application event totals		212.65	14.21	347.51	3,410.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.70	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.70	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.70	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	71.58	5,796,000.00 *gal*
Application event totals		0.00	0.00	0.00	71.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	70.03	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	70.03	

Skiff - 10/29/2022: Oats, silage-soft dough

Field name: Skiff

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

Skiff - 10/29/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/21/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	152.82	44.31	88.51	2,236.28	420.00 *ton*
Application event totals		152.82	44.31	88.51	2,236.28	

09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	0.00	0.00	0.00	170.27	4,290,000.00 *gal*
Application event totals		0.00	0.00	0.00	170.27	

Skiff - 05/31/2023: Corn, silage

Field name: Skiff

Crop: Corn, silage

Plant date: 05/31/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	282.92	135.31	730.36	6,903.87	570.00 *ton*
Application event totals		282.92	135.31	730.36	6,903.87	

05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	81.60	5,418,000.00 *gal*
Application event totals		0.00	0.00	0.00	81.60	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Skiff - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	77.81	5,166,000.00 *gal*
Application event totals		0.00	0.00	0.00	77.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	77.81	5,166,000.00 *gal*
Application event totals		0.00	0.00	0.00	77.81	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	75.91	5,040,000.00 *gal*
Application event totals		0.00	0.00	0.00	75.91	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.01	4,914,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.01	4,914,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.01	

Skiff - 05/31/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	72.12	4,788,000.00 *gal*
Application event totals		0.00	0.00	0.00	72.12	

The 40 - 10/31/2022: Oats, silage-soft dough

Field name: The 40

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/03/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	0.00	0.00	0.00	173.89	2,244,000.00 *gal*
Application event totals		0.00	0.00	0.00	173.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	194.68	45.01	331.07	1,779.77	858,000.00 *gal*
Application event totals		194.68	45.01	331.07	1,779.77	

The 40 - 06/06/2023: Corn, silage

Field name: The 40

Crop: Corn, silage

Plant date: 06/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

The 40 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/05/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	290.72	139.04	750.50	7,094.20	300.00 *ton*
Application event totals		290.72	139.04	750.50	7,094.20	

05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	81.51	2,772,000.00 *gal*
Application event totals		0.00	0.00	0.00	81.51	

06/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.10	2,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.10	

06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	77.81	2,646,000.00 *gal*
Application event totals		0.00	0.00	0.00	77.81	

The 40 - 06/06/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.10	2,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.10	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.10	2,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.10	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	70.40	2,394,000.00 *gal*
Application event totals		0.00	0.00	0.00	70.40	

09/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	66.69	2,268,000.00 *gal*
Application event totals		0.00	0.00	0.00	66.69	

The 80 - 11/01/2022: Oats, silage-soft dough

Field name: The 80

Crop: Oats, silage-soft dough

Plant date: 11/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	0.90	0.00	0.00	98.80	3,498,000.00 *gal*
Application event totals		0.90	0.00	0.00	98.80	

The 80 - 11/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	198.95	31.71	354.33	2,667.59	3,498,000.00 *gal*
Application event totals		198.95	31.71	354.33	2,667.59	

The 80 - 06/07/2023: Corn, silage

Field name: The 80

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	81.40	8,568,000.00 *gal*
Application event totals		0.00	0.00	0.00	81.40	

06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	79.01	8,316,000.00 *gal*
Application event totals		0.00	0.00	0.00	79.01	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	135.74	33.55	220.39	1,471.83	2,970,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	53.87	5,670,000.00 *gal*
Application event totals		135.74	33.55	220.39	1,525.69	

The 80 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	76.61	8,064,000.00 *gal*
Application event totals		0.00	0.00	0.00	76.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	132.73	32.81	215.50	1,439.12	2,904,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	52.67	5,544,000.00 *gal*
Application event totals		132.73	32.81	215.50	1,491.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.22	7,812,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.22	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	71.82	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	71.82	

Tules - 11/02/2022: Oats, silage-soft dough

Field name: Tules

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	207.43	47.95	352.74	1,896.29	2,046,000.00 *gal*
Application event totals		207.43	47.95	352.74	1,896.29	

Tules - 11/02/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	0.00	0.00	0.00	173.67	5,016,000.00 *gal*
Application event totals		0.00	0.00	0.00	173.67	

Tules - 06/03/2023: Corn, silage

Field name: Tules

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	81.12	6,174,000.00 *gal*
Application event totals		0.00	0.00	0.00	81.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	79.46	6,048,000.00 *gal*
Application event totals		0.00	0.00	0.00	79.46	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	135.58	33.51	220.13	1,470.09	2,145,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	53.80	4,095,000.00 *gal*
Application event totals		135.58	33.51	220.13	1,523.89	

Tules - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	76.15	5,796,000.00 *gal*
Application event totals		0.00	0.00	0.00	76.15	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	133.50	33.00	216.75	1,447.47	2,112,000.00 *gal*
Canal	Ground water	0.00	0.00	0.00	52.98	4,032,000.00 *gal*
Application event totals		133.50	33.00	216.75	1,500.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	74.50	5,670,000.00 *gal*
Application event totals		0.00	0.00	0.00	74.50	

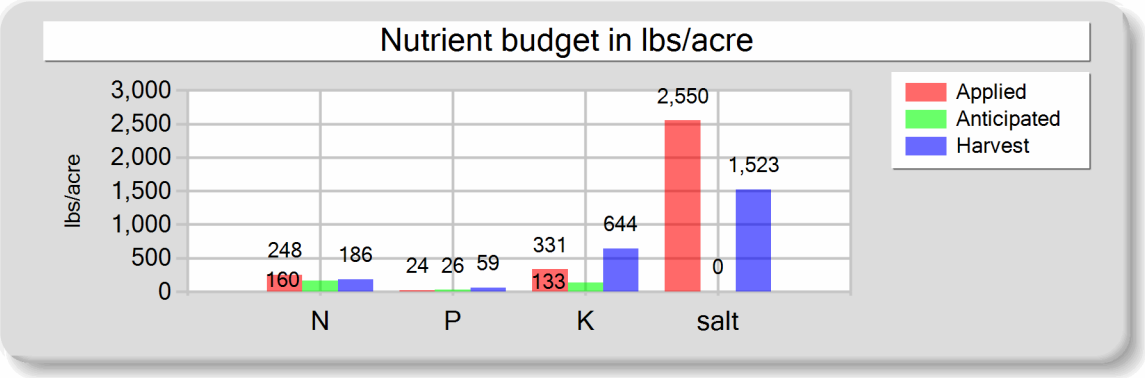
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Ground water	0.00	0.00	0.00	72.84	5,544,000.00 *gal*
Application event totals		0.00	0.00	0.00	72.84	

B. NUTRIENT BUDGET

Big Field - 10/28/2022: Oats, silage-soft dough

Field name: Big Field | Crop: Oats, silage-soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	241.22	23.56	331.07	2,465.72
Fresh water	0.00	0.00	0.00	84.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.22	23.56	331.07	2,549.96
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	185.74	58.53	643.86	1,522.70
Nutrient balance	62.47	-34.97	-312.79	1,027.26
Applied to removed ratio	1.34	0.40	0.51	1.67

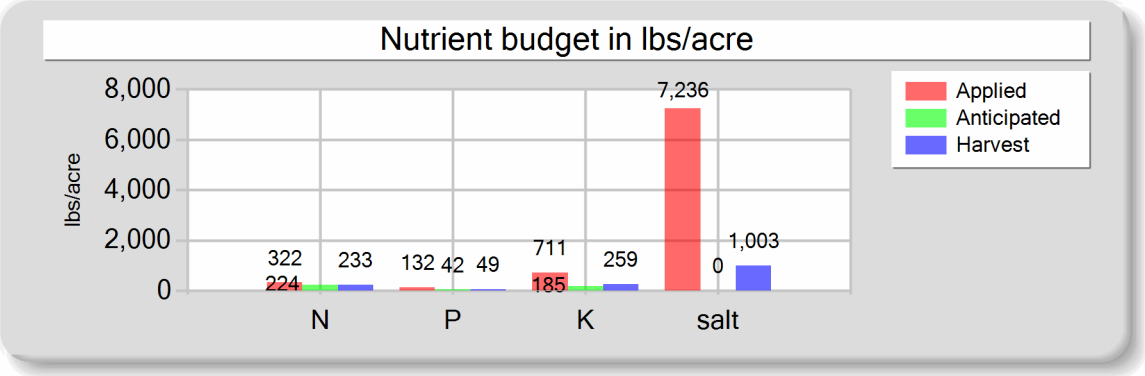
Fresh water applied	
2,640,000.00	*gallons*
97.22	*acre-inches*
1.91	*inches/acre*

Process wastewater applied	
3,696,000.00	*gallons*
136.11	*acre-inches*
2.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

Big Field - 06/05/2023: Corn, silage

Field name: Big Field Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	275.33	131.68	710.77	6,718.62
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	517.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	322.33	131.68	710.77	7,235.82
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	232.73	49.02	259.21	1,003.48
Nutrient balance	89.60	82.65	451.56	6,232.34
Applied to removed ratio	1.39	2.69	2.74	7.21

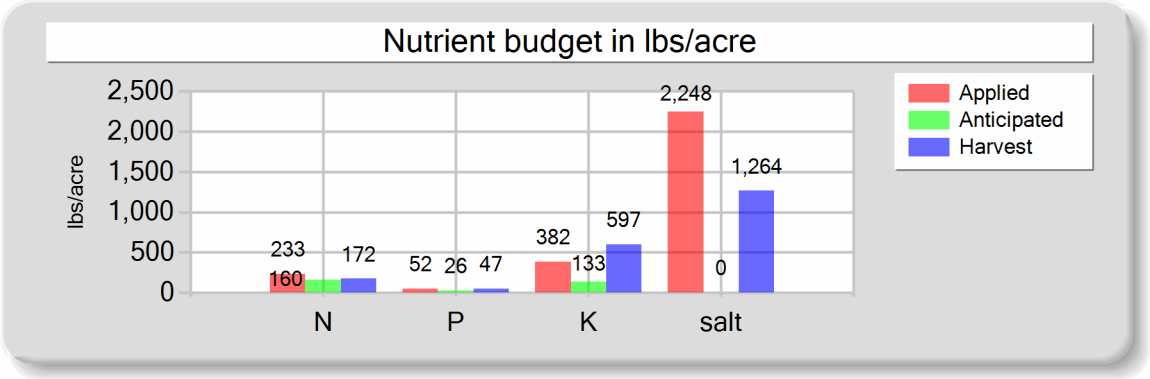
Fresh water applied	
42,714,000.00	*gallons*
1,573.01	*acre-inches*
30.84	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Faria 1 - 10/28/2022: Oats, silage-soft dough

Field name: Faria 1 Crop: Oats, silage-soft dough Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	224.64	51.93	382.00	2,053.59
Fresh water	1.77	0.00	0.00	194.74
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	233.41	51.93	382.00	2,248.32
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	172.08	46.82	597.01	1,264.25
Nutrient balance	61.33	5.11	-215.01	984.07
Applied to removed ratio	1.36	1.11	0.64	1.78

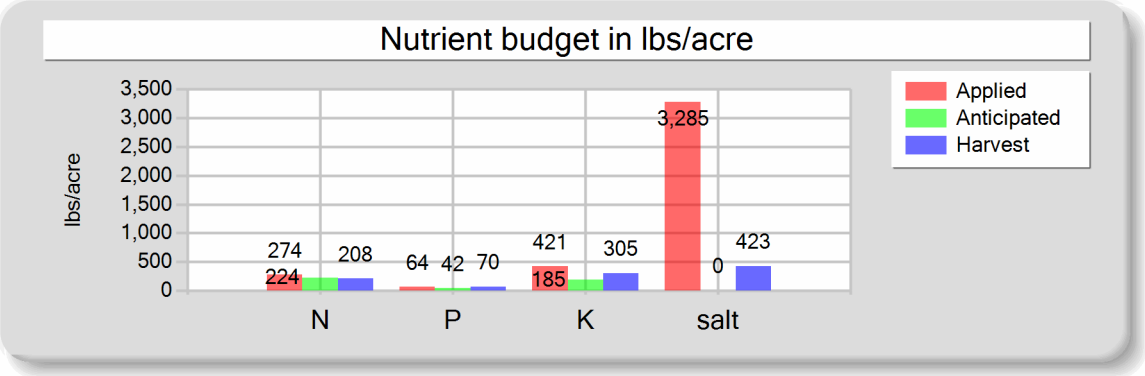
Fresh water applied	
2,970,000.00	*gallons*
109.38	*acre-inches*
3.91	*inches/acre*

Process wastewater applied	
1,320,000.00	*gallons*
48.61	*acre-inches*
1.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Faria 1 - 06/02/2023: Corn, silage

Field name: Faria 1 Crop: Corn, silage Plant date: 06/02/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	259.10	64.05	420.67	2,809.32
Fresh water	0.00	0.00	0.00	475.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	274.10	64.05	420.67	3,284.51
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	207.55	69.94	304.56	422.89
Nutrient balance	66.55	-5.89	116.11	2,861.62
Applied to removed ratio	1.32	0.92	1.38	7.77

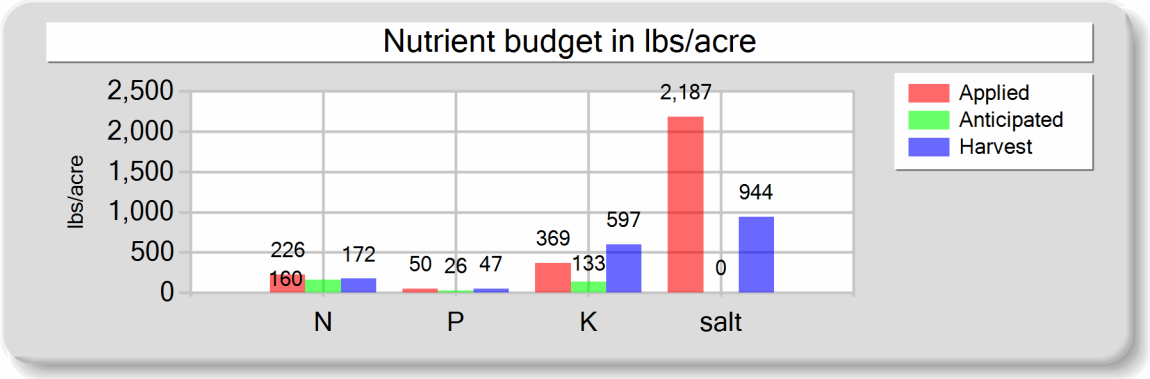
Fresh water applied	
21,546,000.00	*gallons*
793.47	*acre-inches*
28.34	*inches/acre*

Process wastewater applied	
2,442,000.00	*gallons*
89.93	*acre-inches*
3.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

Faria 2 - 10/29/2022: Oats, silage-soft dough

Field name: Faria 2 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	217.15	50.20	369.27	1,985.13
Fresh water	1.84	0.00	0.00	201.95
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.98	50.20	369.27	2,187.08
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	172.08	46.82	597.01	943.97
Nutrient balance	53.91	3.37	-227.74	1,243.11
Applied to removed ratio	1.31	1.07	0.62	2.32

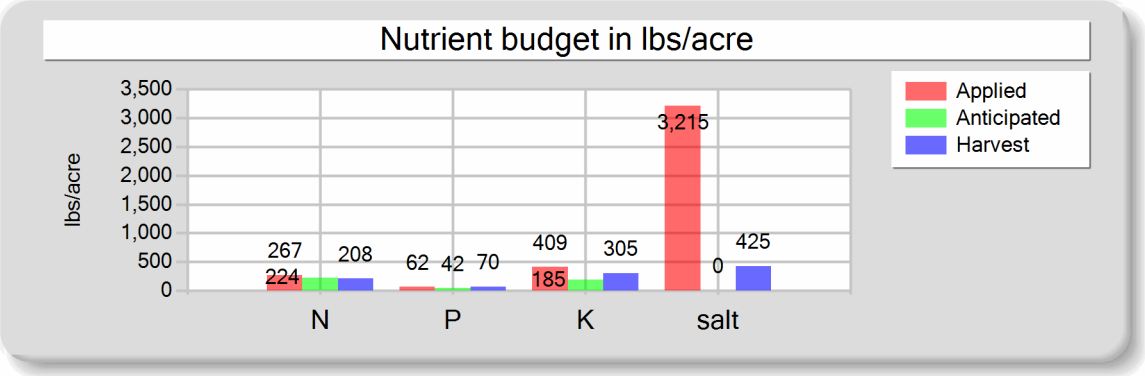
Fresh water applied	
2,310,000.00	*gallons*
85.07	*acre-inches*
4.05	*inches/acre*

Process wastewater applied	
957,000.00	*gallons*
35.24	*acre-inches*
1.68	*inches/acre*

Total harvests for the crop	
1	*harvests*

Faria 2 - 06/04/2023: Corn, silage

Field name: Faria 2 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	252.10	62.32	409.30	2,733.39
Fresh water	0.00	0.00	0.00	481.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.10	62.32	409.30	3,215.07
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	207.55	69.94	304.56	425.48
Nutrient balance	59.54	-7.62	104.74	2,789.58
Applied to removed ratio	1.29	0.89	1.34	7.56

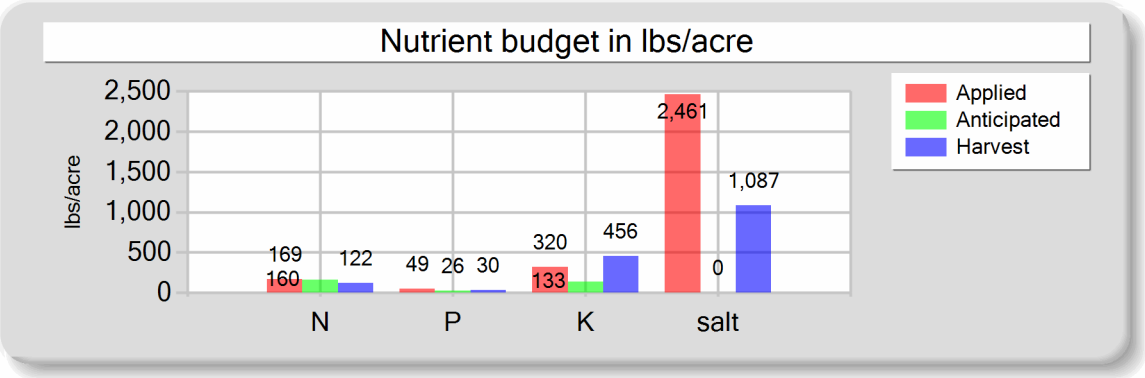
Fresh water applied	
16,380,000.00	*gallons*
603.22	*acre-inches*
28.72	*inches/acre*

Process wastewater applied	
1,782,000.00	*gallons*
65.63	*acre-inches*
3.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morris - 10/30/2022: Oats, silage-soft dough

Field name: Morris | Crop: Oats, silage-soft dough | Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	80.99	38.72	159.84	962.59
Process wastewater	81.43	9.91	160.47	1,321.85
Fresh water	0.00	0.00	0.00	176.14
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.41	48.63	320.31	2,460.58
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	122.00	30.41	456.19	1,087.40
Nutrient balance	47.41	18.22	-135.88	1,373.18
Applied to removed ratio	1.39	1.60	0.70	2.26

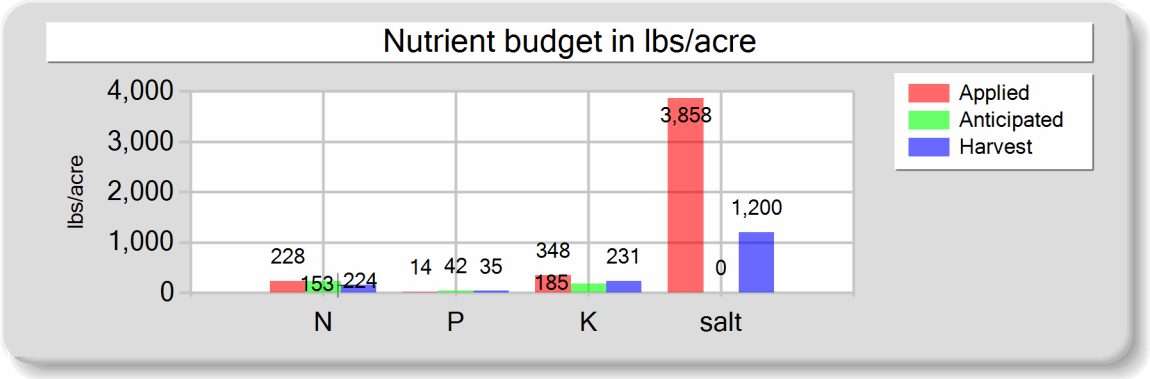
Fresh water applied	
5,148,000.00	*gallons*
189.58	*acre-inches*
3.79	*inches/acre*

Process wastewater applied	
1,584,000.00	*gallons*
58.33	*acre-inches*
1.17	*inches/acre*

Total harvests for the crop	
1	*harvests*

Morris - 06/11/2023: Corn, silage

Field name: Morris | Crop: Corn, silage | Plant date: 06/11/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.65	14.21	347.51	3,358.82
Fresh water	0.00	0.00	0.00	499.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.65	14.21	347.51	3,858.35
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	153.04	35.19	231.20	1,200.43
Nutrient balance	74.61	-20.98	116.32	2,657.92
Applied to removed ratio	1.49	0.40	1.50	3.21

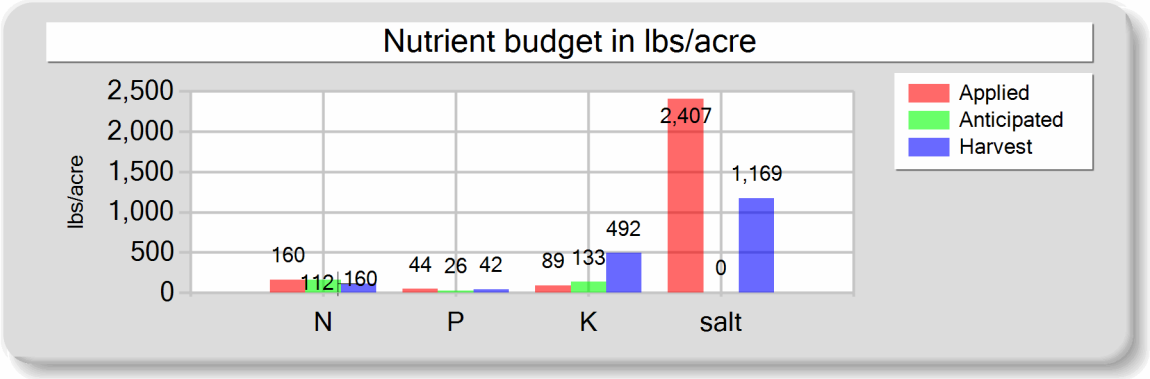
Fresh water applied	
40,446,000.00	*gallons*
1,489.49	*acre-inches*
29.79	*inches/acre*

Process wastewater applied	
2,178,000.00	*gallons*
80.21	*acre-inches*
1.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Skiff - 10/29/2022: Oats, silage-soft dough

Field name: Skiff | Crop: Oats, silage-soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	152.82	44.31	88.51	2,236.28
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	170.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	159.82	44.31	88.51	2,406.55
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	112.38	42.14	491.65	1,169.43
Nutrient balance	47.45	2.17	-403.14	1,237.12
Applied to removed ratio	1.42	1.05	0.18	2.06

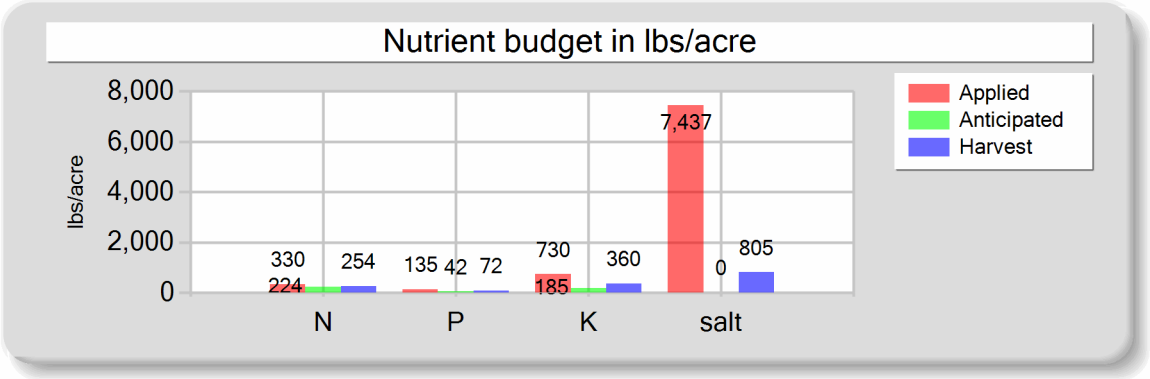
Fresh water applied	
4,290,000.00	*gallons*
157.99	*acre-inches*
3.85	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Skiff - 05/31/2023: Corn, silage

Field name: Skiff | Crop: Corn, silage | Plant date: 05/31/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	282.92	135.31	730.36	6,903.87
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	533.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	329.92	135.31	730.36	7,437.14
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	253.80	71.63	360.40	805.05
Nutrient balance	76.12	63.68	369.97	6,632.09
Applied to removed ratio	1.30	1.89	2.03	9.24

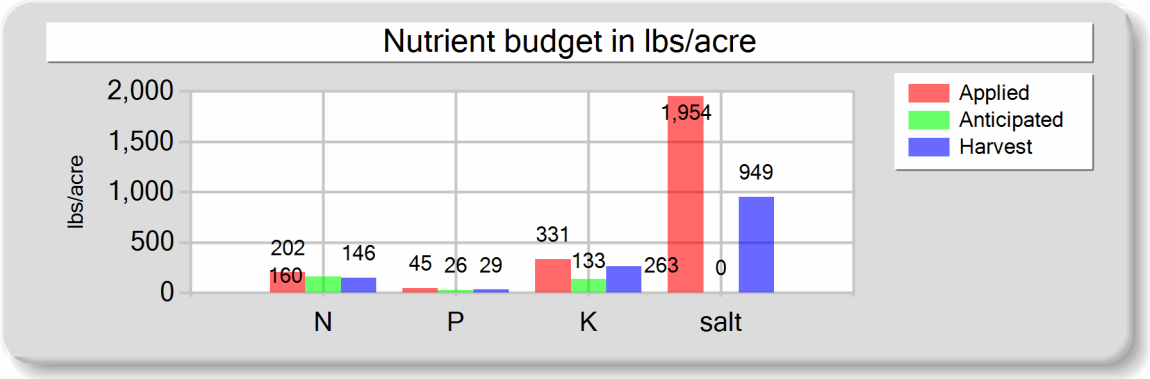
Fresh water applied	
35,406,000.00	*gallons*
1,303.88	*acre-inches*
31.80	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 40 - 10/31/2022: Oats, silage-soft dough

Field name: The 40 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	194.68	45.01	331.07	1,779.77
Fresh water	0.00	0.00	0.00	173.89
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.68	45.01	331.07	1,953.66
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	145.93	29.27	262.99	949.43
Nutrient balance	55.75	15.74	68.07	1,004.23
Applied to removed ratio	1.38	1.54	1.26	2.06

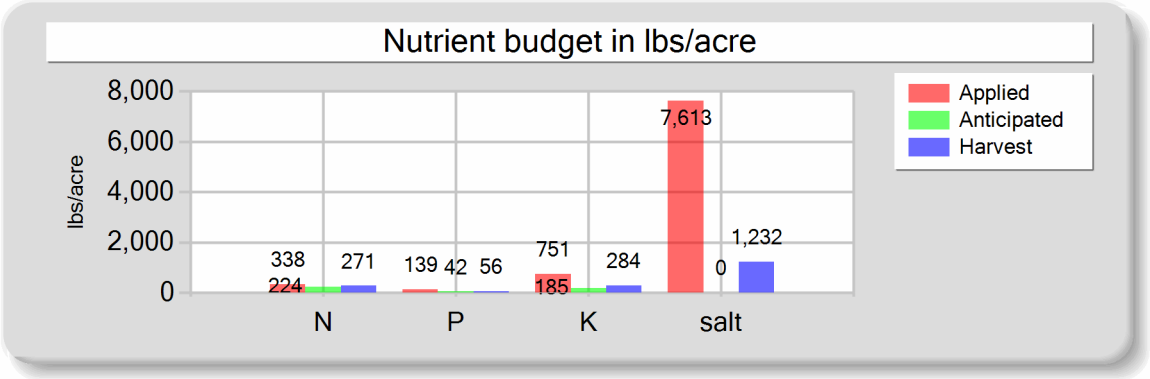
Fresh water applied	
2,244,000.00	*gallons*
82.64	*acre-inches*
3.94	*inches/acre*

Process wastewater applied	
858,000.00	*gallons*
31.60	*acre-inches*
1.50	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 40 - 06/06/2023: Corn, silage

Field name: The 40 Crop: Corn, silage Plant date: 06/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	290.72	139.04	750.50	7,094.20
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	518.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	337.72	139.04	750.50	7,612.93
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	270.72	55.84	283.69	1,231.78
Nutrient balance	67.00	83.20	466.81	6,381.15
Applied to removed ratio	1.25	2.49	2.65	6.18

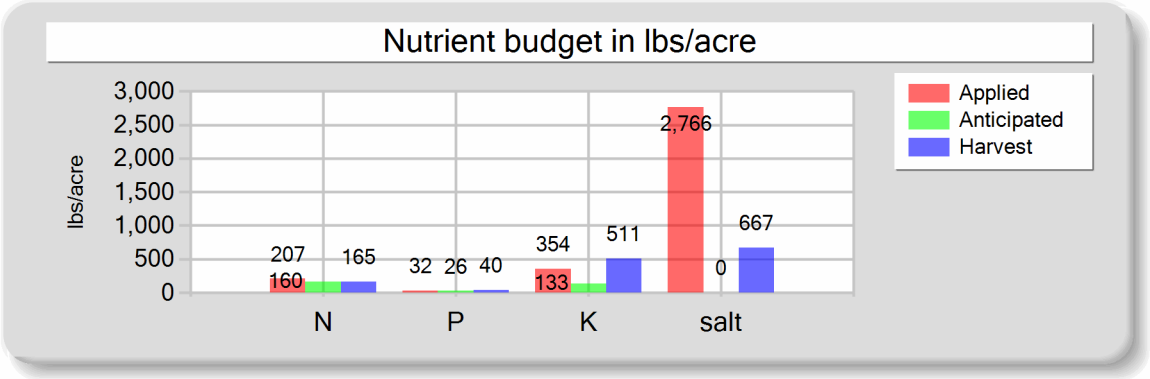
Fresh water applied	
17,640,000.00	*gallons*
649.62	*acre-inches*
30.93	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 80 - 11/01/2022: Oats, silage-soft dough

Field name: The 80 Crop: Oats, silage-soft dough Plant date: 11/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	198.95	31.71	354.33	2,667.59
Fresh water	0.90	0.00	0.00	98.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	206.84	31.71	354.33	2,766.39
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	165.44	40.19	511.16	667.45
Nutrient balance	41.40	-8.48	-156.83	2,098.94
Applied to removed ratio	1.25	0.79	0.69	4.14

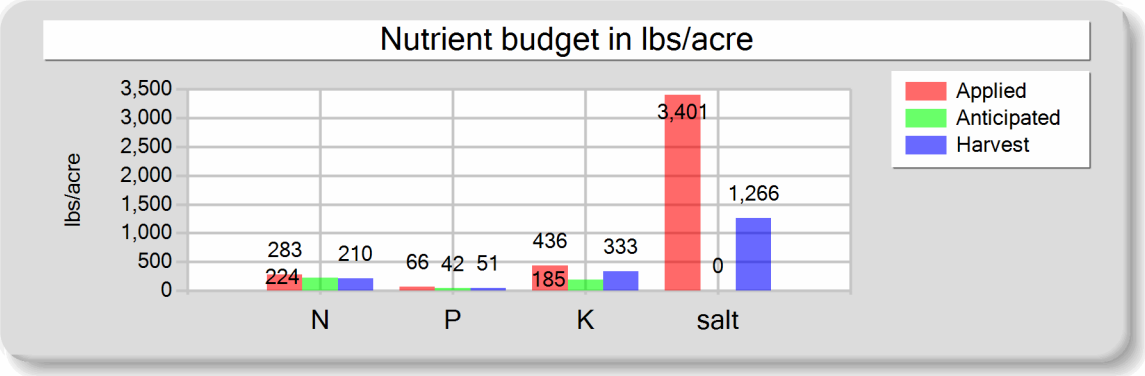
Fresh water applied
3,498,000.00 *gallons*
128.82 *acre-inches*
1.98 *inches/acre*

Process wastewater applied
3,498,000.00 *gallons*
128.82 *acre-inches*
1.98 *inches/acre*

Total harvests for the crop
1 *harvests*

The 80 - 06/07/2023: Corn, silage

Field name: The 80 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	268.47	66.36	435.89	2,910.95
Fresh water	0.00	0.00	0.00	489.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	283.47	66.36	435.89	3,400.54
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	210.37	50.76	332.76	1,265.62
Nutrient balance	73.10	15.60	103.13	2,134.93
Applied to removed ratio	1.35	1.31	1.31	2.69

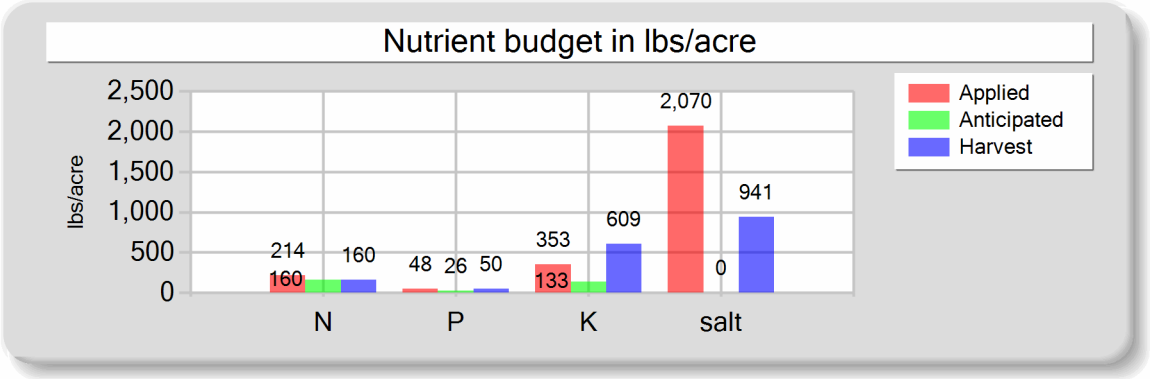
Fresh water applied	
51,534,000.00	*gallons*
1,897.82	*acre-inches*
29.20	*inches/acre*

Process wastewater applied	
5,874,000.00	*gallons*
216.32	*acre-inches*
3.33	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tules - 11/02/2022: Oats, silage-soft dough

Field name: Tules | Crop: Oats, silage-soft dough | Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.43	47.95	352.74	1,896.29
Fresh water	0.00	0.00	0.00	173.67
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	214.43	47.95	352.74	2,069.96
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	160.37	50.34	608.71	941.47
Nutrient balance	54.06	-2.38	-255.97	1,128.48
Applied to removed ratio	1.34	0.95	0.58	2.20

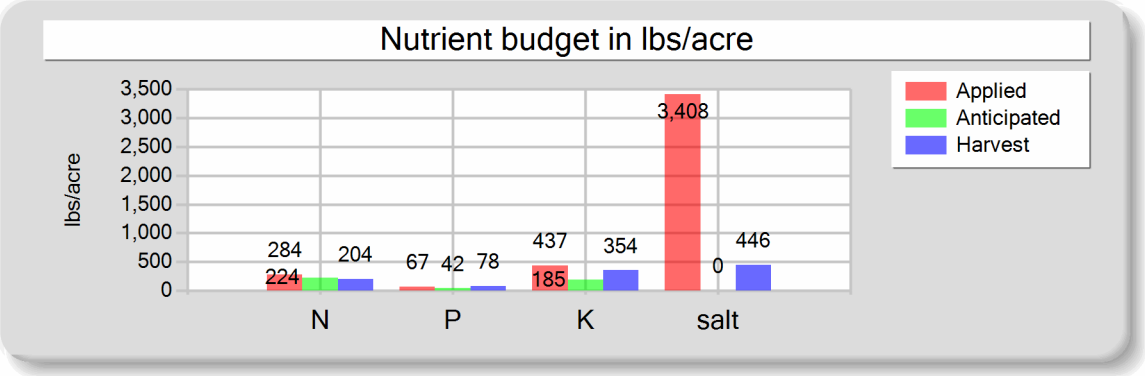
Fresh water applied	
5,016,000.00	*gallons*
184.72	*acre-inches*
3.93	*inches/acre*

Process wastewater applied	
2,046,000.00	*gallons*
75.35	*acre-inches*
1.60	*inches/acre*

Total harvests for the crop	
1	*harvests*

Tules - 06/03/2023: Corn, silage

Field name: Tules | Crop: Corn, silage | Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	269.08	66.51	436.88	2,917.56
Fresh water	0.00	0.00	0.00	490.86
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	284.08	66.51	436.88	3,408.42
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	204.17	78.40	354.19	445.79
Nutrient balance	79.91	-11.88	82.69	2,962.63
Applied to removed ratio	1.39	0.85	1.23	7.65

Fresh water applied	
37,359,000.00	*gallons*
1,375.80	*acre-inches*
29.27	*inches/acre*

Process wastewater applied	
4,257,000.00	*gallons*
156.77	*acre-inches*
3.34	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,900.00	7,510.00	15,000.00	29,400.00	8,480.00	2,120.00	0.00	1,800.00		37.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,800.00	10,900.00	45,000.00							27.10
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,400.00	8,800.00	47,500.00							44.90
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 32.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	6,500.00	46,000.00							48.20
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 52.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	4,720.00	12,700.00							22.70
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	6,110.00	31,100.00							29.80
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

3QWW

Sample and source description: 3QWW

Sample date: 08/08/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	330.00	123.00	0.00	0.00	1.75	378.00								5,410.00	3,620
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/24/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	443.00	310.00	0.00	0.00	70.60	789.00								6,100.00	5,940
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	571.00	316.00	0.00	0.00	132.00	971.00								6,700.00	5,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	426.00	132.00	0.00	0.00	73.30	654.00								5,970.00	3,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	680.00	249.00	0.00	0.00	68.60	495.00								3,970.00	1,960
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW Heifer

Sample and source description: 2QWW Heifer

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	308.00	185.00	0.00	0.00	37.50	607.00								6,110.00	5,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	356.00	258.00	0.00	0.00	88.00	578.00								5,170.00	3,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW Heifer

Sample and source description: 3QWW Heifer

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	585.00	426.00	0.00	0.00	39.10	956.00								8,550.00	9,240
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	482.00	311.00	0.00	0.00	75.00	829.00								6,300.00	4,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW Heifer

Sample and source description: 4QWW Heifer

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	291.00	101.00	0.00	0.00	51.40	638.00								4,020.00	5,860
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal

Fresh Water

Sample description: Fresh Water

Sample date: 09/28/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								78.00	74
DL	0.50	0.50	0.50								1.00	10

IW 1

IW 1

Fresh Water

Sample description: Fresh Water

Sample date: 09/28/2022 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								218.00	195
DL	0.50	0.50	0.50								1.00	10

Fresh Water

Sample description: Fresh Water

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.78	0.00	2.78								300.00	2,160
DL	0.50	0.50	0.50								1.00	10

IW 2

Fresh Water

Sample description: Fresh Water

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.90	0.00	2.90								363.00	242
DL	0.50	0.50	0.50								1.00	10

IW 3

IW 3

Fresh Water

Sample description: Fresh Water

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.64	0.00	1.60								232.00	188
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

Big Field

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			98.00				
DL			4.00				

Faria 1

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			96.00				
DL			4.00				

Faria 2

Faria 2

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			96.00				
DL			4.00				

Morris

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			121.00				
DL			4.00				

Skiff

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			4.00				

The 40

The 40

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			98.00				
DL			4.00				

The 80

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			88.00				
DL			4.00				

Tules

Soils

Sample and source description: Soils

Sample date: 12/05/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

Big Field - 10/28/2022: Oats, silage-soft dough

Big Field - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,760.00	1,500.00	16,500.00		10.90
DL	100.00	100.00	100.00		0.01

Big Field - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,130.00	870.00	4,600.00		5.30
DL	100.00	100.00	100.00		0.01

Faria 1 - 10/28/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,410.00	1,200.00	15,300.00		10.80
DL	100.00	100.00	100.00		0.01

Faria 1 - 06/02/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,680.00	1,240.00	5,400.00		2.30
DL	100.00	100.00	100.00		0.01

Faria 2 - 10/29/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 77.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,410.00	1,200.00	15,300.00		10.80
DL	100.00	100.00	100.00		0.01

Morris - 10/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,530.00	880.00	13,200.00		11.40
DL	100.00	100.00	100.00		0.01

Morris - 06/11/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	860.00	5,650.00		7.60
DL	100.00	100.00	100.00		0.01

Skiff - 10/29/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,880.00	1,080.00	12,600.00		9.99
DL	100.00	100.00	100.00		0.01

Skiff - 05/31/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,500.00	1,270.00	6,390.00		3.90
DL	100.00	100.00	100.00		0.01

The 40 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,740.00	750.00	6,740.00		8.69
DL	100.00	100.00	100.00		0.01

The 40 - 06/06/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,800.00	990.00	5,030.00		6.50
DL	100.00	100.00	100.00		0.01

The 80 - 11/01/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 76.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,240.00	1,030.00	13,100.00		7.31
DL	100.00	100.00	100.00		0.01

The 80 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,730.00	900.00	5,900.00		6.80
DL	100.00	100.00	100.00		0.01

Tules - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/09/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,110.00	1,290.00	15,600.00		11.60
DL	100.00	100.00	100.00		0.01

Tules - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/02/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,620.00	1,390.00	6,280.00		2.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

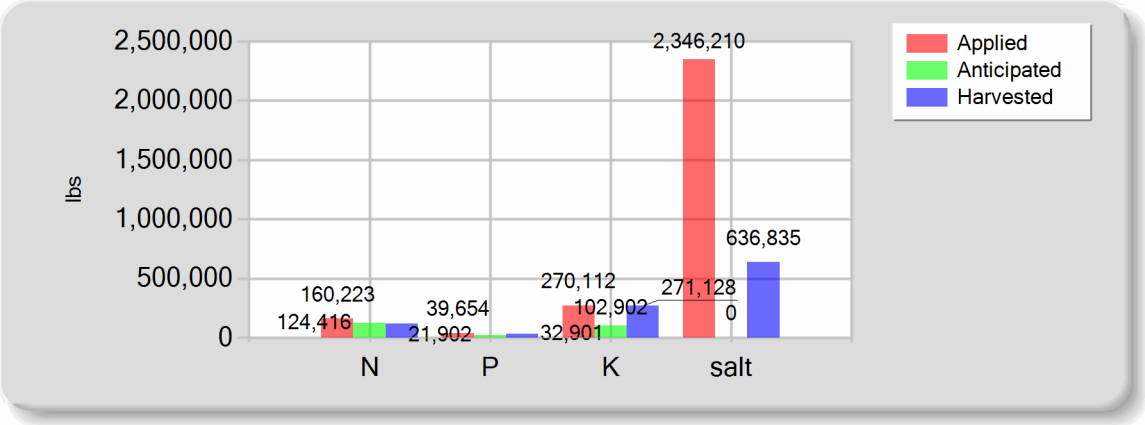
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

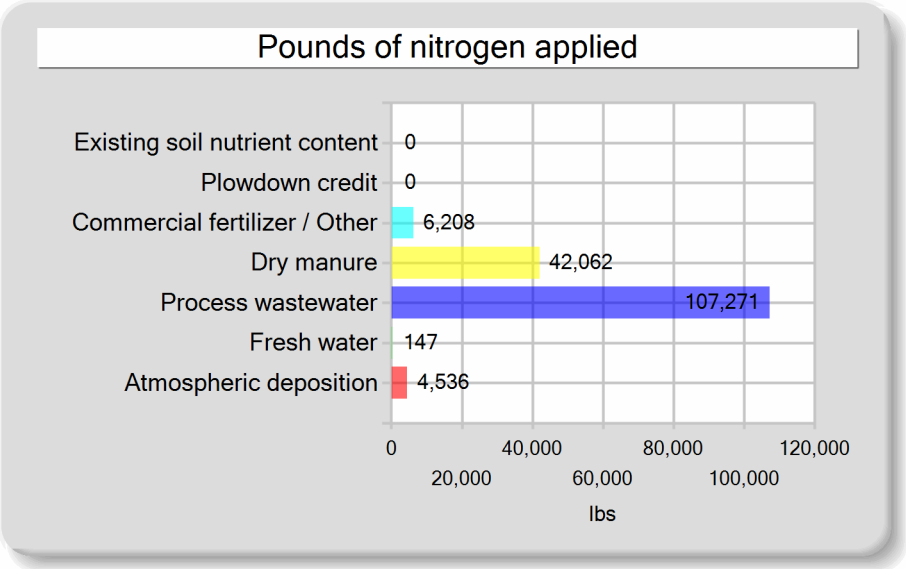
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

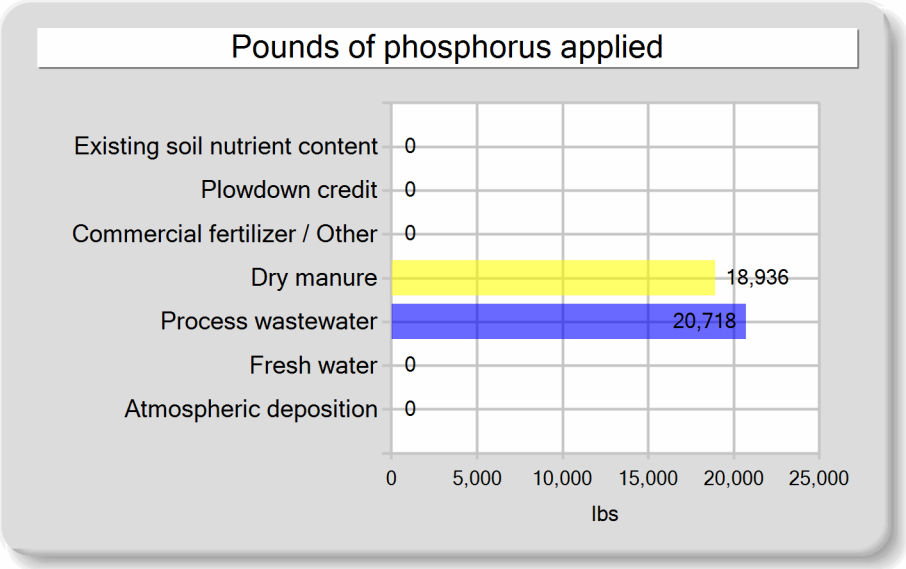
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	6,208.00	0.00	0.00	0.00
Dry manure	42,061.63	18,935.83	93,575.40	914,503.92
Process wastewater	107,270.99	20,717.82	176,536.93	1,221,266.38
Fresh water	146.50	0.00	0.00	210,439.20
Atmospheric deposition	4,536.00	0.00	0.00	0.00
Total nutrients applied	160,223.13	39,653.65	270,112.33	2,346,209.51
Anticipated crop nutrient removal	124,416.00	21,902.40	102,902.40	0.00
Actual crop nutrient removal	119,020.19	32,900.71	271,128.45	636,834.86
Nutrient balance	41,202.94	6,752.94	-1,016.12	1,709,374.65
Applied to removed ratio	1.35	1.21	1.00	3.68

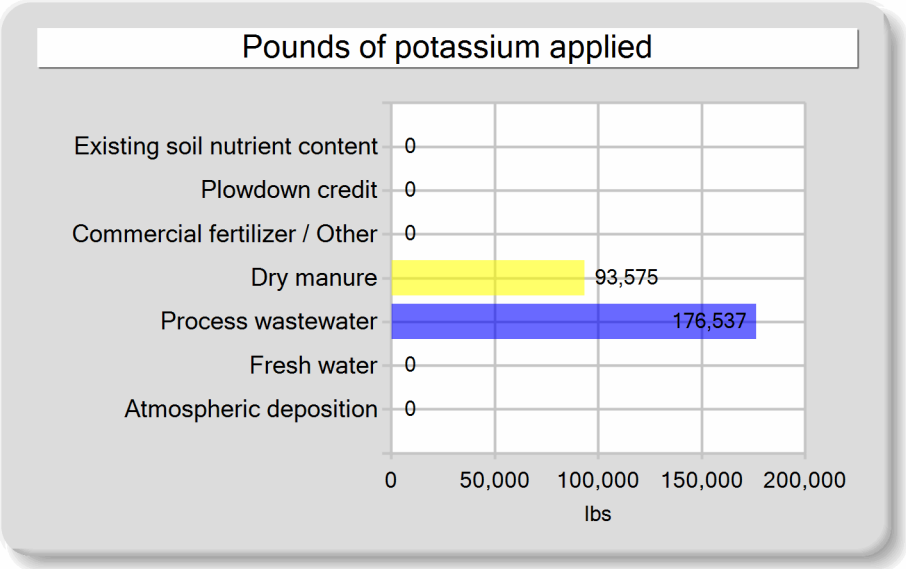
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

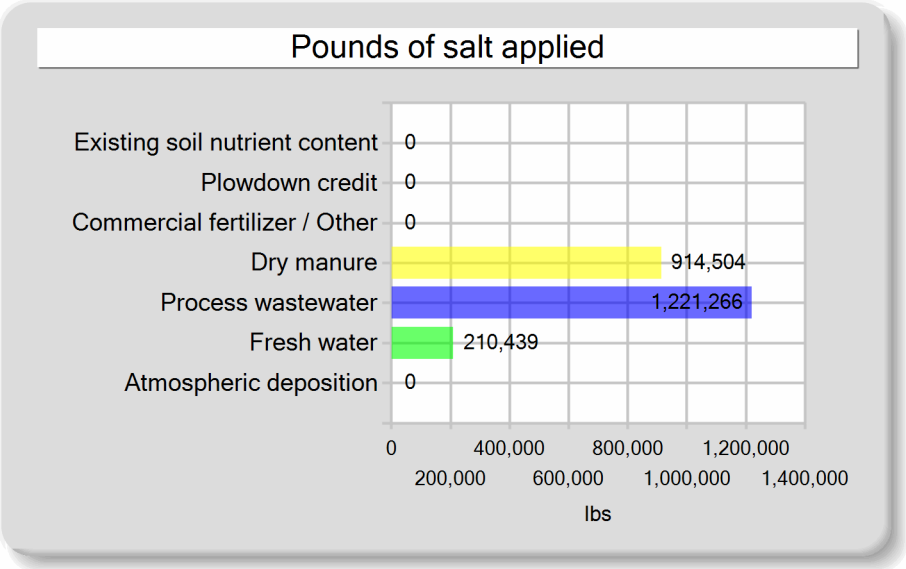


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Morris Corn (1.49) had high nitrogen field ratio as Field failed to meet anticipated nitrogen removal of 224 pounds/acre. The subject fields removed 153 pounds of nitrogen per acre. Yield for the field was 20.5 tons per acre, well below the anticipated 28 tons per acre. Also, a low nitrogen lab value from the plant tissue lab analysis resulted in 7.48 pounds of nitrogen removed per harvested ton compared to the anticipated 8 pounds of nitrogen removed per harvested ton. These two factors were the primary reason for the high nitrogen field ratio. Nitrogen application to the fields was not considered excessive but in-line with that recommended in the dairy's Nutrient Management Plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Manuel Borges
Manuel Borges (Jun 25, 2024 06:10 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Manuel Borges	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 25, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0954-01	IW 1	Canal Water			12/13/2023 7:00
23L0954-02	IW 2	Canal Water			12/13/2023 7:00
23L0954-03	IW 3	Canal Water			12/13/2023 7:00
23L0954-04	DW 1 Heifer	Well Water			12/13/2023 7:00
23L0954-05	DW 2 Heifer	Well Water			12/13/2023 7:00
23L0954-06	DW House East	Well Water			12/13/2023 7:00
23L0954-07	DW Milk Barn	Well Water			12/13/2023 7:00
23L0954-08	DW Skiff	Well Water			12/13/2023 7:00

Default Cooler Temperature on Receipt °C: 1.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Sample Results

Sample: IW 1
23L0954-01 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.30**	mmhos/cm	0.01	1		12/18/23 14:45	SM 2510 B		BEL0679
Electrical Conductivity umhos	**300**	umhos/cm	10.0	1		12/18/23 14:45	SM 2510 B		BEL0679
Nitrate Nitrogen as NO3N	**2.8**	mg/L	0.1	1	10	12/15/23 07:23	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 14:45	SM 4500-H+	H	BEL0679
pH	**7.7**	units	1.0	1		12/18/23 14:45	SM 4500-H+	H	BEL0679
Total Filterable Solids (TDS)	**2160**	mg/L	10.0	1		12/18/23 17:07	SM 2540 C		BEL0601
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		12/19/23 14:05	SM 4500-NH3 C		BEL0671
Total Nitrogen	**2.78**	mg/L	1.00	1		12/19/23 14:05	SM 4500-NH3 C		BEL0671

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Sample Results
(Continued)

Sample: IW 2
23L0954-02 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.36**	mmhos/cm	0.01	1		12/18/23 14:46	SM 2510 B		BEL0679
Electrical Conductivity umhos	**363**	umhos/cm	10.0	1		12/18/23 14:46	SM 2510 B		BEL0679
Nitrate Nitrogen as NO3N	**2.9**	mg/L	0.1	1	10	12/15/23 07:45	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 14:46	SM 4500-H+	H	BEL0679
pH	**8.1**	units	1.0	1		12/18/23 14:46	SM 4500-H+	H	BEL0679
Total Filterable Solids (TDS)	**242**	mg/L	10.0	1		12/18/23 17:07	SM 2540 C		BEL0601
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		12/19/23 14:07	SM 4500-NH3 C		BEL0671
Total Nitrogen	**2.90**	mg/L	1.00	1		12/19/23 14:07	SM 4500-NH3 C		BEL0671

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Sample Results
(Continued)

Sample: IW 3
23L0954-03 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.23**	mmhos/cm	0.01	1		12/18/23 15:16	SM 2510 B		BEL0680
Electrical Conductivity umhos	**232**	umhos/cm	10.0	1		12/18/23 15:16	SM 2510 B		BEL0680
Nitrate Nitrogen as NO3N	**1.6**	mg/L	0.1	1	10	12/15/23 08:06	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 15:16	SM 4500-H+	H	BEL0680
pH	**7.1**	units	1.0	1		12/18/23 15:16	SM 4500-H+	H	BEL0680
Total Filterable Solids (TDS)	**188**	mg/L	10.0	1		12/18/23 17:07	SM 2540 C		BEL0601
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		12/19/23 14:08	SM 4500-NH3 C		BEL0671
Total Nitrogen	**1.64**	mg/L	1.00	1		12/19/23 14:08	SM 4500-NH3 C		BEL0671

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Sample Results
(Continued)

Sample: DW 1 Heifer
23L0954-04 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.31**	mmhos/cm	0.01	1		12/18/23 15:18	SM 2510 B		BEL0680
Electrical Conductivity umhos	**314**	umhos/cm	10.0	1		12/18/23 15:18	SM 2510 B		BEL0680
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**2.5**	mg/L	0.1	1	10	12/15/23 08:29	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 15:18	SM 4500-H+	H	BEL0680
pH	**7.6**	units	1.0	1		12/18/23 15:18	SM 4500-H+	H	BEL0680

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Sample Results
(Continued)

Sample: DW 2 Heifer
23L0954-05 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.30**	mmhos/cm	0.01	1		12/18/23 15:20	SM 2510 B		BEL0680
Electrical Conductivity umhos	**295**	umhos/cm	10.0	1		12/18/23 15:20	SM 2510 B		BEL0680
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**2.2**	mg/L	0.1	1	10	12/15/23 11:24	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 15:20	SM 4500-H+	H	BEL0680
pH	**7.8**	units	1.0	1		12/18/23 15:20	SM 4500-H+	H	BEL0680

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Sample Results
(Continued)

Sample: DW House East
23L0954-06 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.23**	mmhos/cm	0.01	1		12/18/23 15:21	SM 2510 B		BEL0680
Electrical Conductivity umhos	**230**	umhos/cm	10.0	1		12/18/23 15:21	SM 2510 B		BEL0680
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**1.6**	mg/L	0.1	1	10	12/15/23 11:46	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 15:21	SM 4500-H+	H	BEL0680
pH	**8.1**	units	1.0	1		12/18/23 15:21	SM 4500-H+	H	BEL0680

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Sample Results
(Continued)

Sample: DW Milk Barn
23L0954-07 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.38**	mmhos/cm	0.01	1		12/18/23 15:23	SM 2510 B		BEL0680
Electrical Conductivity umhos	**384**	umhos/cm	10.0	1		12/18/23 15:23	SM 2510 B		BEL0680
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**2.3**	mg/L	0.1	1	10	12/15/23 12:08	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 15:23	SM 4500-H+	H	BEL0680
pH	**8.0**	units	1.0	1		12/18/23 15:23	SM 4500-H+	H	BEL0680

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Manuel & Jeanette Borges
10408 S Van Allen Rd
Stockton, CA 95215

Account# 00-0026361
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 12/14/2023 14:20
Reported: 12/21/2023 10:10

Sample Results
(Continued)

Sample: DW Skiff
23L0954-08 (Water)

Sampled: 12/13/2023 7:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.23**	mmhos/cm	0.01	1		12/18/23 15:24	SM 2510 B		BEL0680
Electrical Conductivity umhos	**234**	umhos/cm	10.0	1		12/18/23 15:24	SM 2510 B		BEL0680
Ammonia (as N)	*	mg/L	0.00	1		12/14/23 17:35	Field		BEL0647
Nitrate Nitrogen as NO3N	**1.6**	mg/L	0.1	1	10	12/15/23 12:30	EPA 300.0		BEL0597
Temperature	**25.0**	units	0.0	1		12/18/23 15:24	SM 4500-H+	H	BEL0680
pH	**8.2**	units	1.0	1		12/18/23 15:24	SM 4500-H+	H	BEL0680

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

12/14/23 14:20 23L0954

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

No. Samples: 8 No of Bottles: 11

Purchase Order No

Bill To: Acct # 0 Cons # 13
26361

Results Need By

Name: Manuel & Jeanette Borges
Address: 10408 S Van Allen Rd
City: Stockton State: CA Zip: 95215
Telephone: 209 466-1828 Fax:
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Other:

Analysis and Bottles Required: (Please indicate Analysis)
DCW1 + TN
DWW1

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW 1	12/13/23	started 7 AM	1.6	
2	2			3.9	
3	3			8.8	
4	DW 1 Heifer			11.1	
5	DW 2 Heifer			9.1	
6	DW House East			8.1	
7	DW milk Barn			9.9	
8	DW Skiff			1.8	
9					
10					

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JCF	DLV	12-14-23 10:00	12-14-23 2:15pm
Third				
Fourth		DLV	12/14/23 14:20	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

pH Strips
Lot 10BDH4501
Exp. Jan. 2025

2023 Annual Report Manuel & Jeanette Borges

Final Audit Report 2024-06-25

Created:	2024-06-24
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAq6WqzWrz5K03D6qkrsiCm-Y9I4QVmV2m

"2023 Annual Report Manuel & Jeanette Borges" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-06-24 - 10:08:38 PM GMT

Document emailed to jborges04@yahoo.com for signature
2024-06-24 - 10:11:02 PM GMT

Email viewed by jborges04@yahoo.com
2024-06-25 - 4:33:51 AM GMT

Signer jborges04@yahoo.com entered name at signing as Manuel Borges
2024-06-25 - 1:10:42 PM GMT

Document e-signed by Manuel Borges (jborges04@yahoo.com)
Signature Date: 2024-06-25 - 1:10:44 PM GMT - Time Source: server

Agreement completed.
2024-06-25 - 1:10:44 PM GMT

Adobe Acrobat Sign